

5 Interesting Reference Designs FOR SMART HOMES

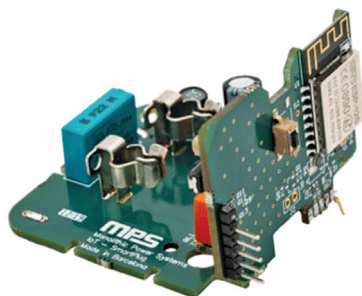
Smart home technology is redefining everyday living by enabling seamless control of devices, energy savings, extended battery life, and intelligent automation, all without major changes to the home.

NIDHI AGARWAL

Smart home devices are transforming the way people interact with their appliances. They make homes more convenient, secure, and energy-efficient. From smart plugs and energy monitors to smart locks and thermostats, reference designs help design engineers create connected products that are easy to use, consume less power, and are reliable. These designs allow you to control devices remotely, track energy use, extend battery life, and automate routines. They provide practical solutions for upgrading homes and small commercial spaces without major modifications.

Non-isolated smart plug

A smart plug can be inserted into a traditional electrical outlet, enabling you to control the power supply to connected devices remotely



through your phone or tablet. This reference design does exactly that by fitting into a standard wall outlet and allowing

remote control, scheduling, and energy monitoring of devices such as lamps, televisions, coffee makers, or fans via Wi-Fi using a smart-phone app or voice assistants like Amazon Alexa or Google Assistant.

The design consists of two boards, one for the DC/DC converter and relay, and the other for the Wi-Fi module that adds IoT capability to non-intelligent devices. This smart plug is an affordable way to upgrade existing setups and automate non-smart devices at home without major changes.

Applications. You can use this design to control devices like lights, fans, heaters, or coffee makers from your phone or through

voice commands. It helps you monitor and save energy, set timers for automatic on/off, and enhance home safety.

Link. <https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-non-isolated-smart-plug>

OEM Brand. Monolithic Power System (MPS)

Wireless energy monitoring system

Smart home products often struggle to track energy use accurately while staying compact, power-efficient, and affordable, as measuring voltage and current, storing data, and sending it to the cloud often require multiple chips and complex firmware. This reference design addresses that challenge with a wireless energy monitor that integrates accurate measurement, Wi-Fi connectivity, and low-power operation in a compact board, enabling real-time energy tracking and remote on/off control.



A 16-bit microcontroller manages the system with an energy monitor sensor, while a Wi-Fi module handles communication and an RTC with battery backup plus 32Mbit flash handles timekeeping and data storage. The design suits smart plugs, connected appliances, building automation systems, and IoT gateways for residential and small commercial energy management.

Power is handled by linear and switching regulators with very low standby consumption, ensuring minimal impact on overall energy use.

Applications. You can use this reference design in smart plugs to track and control